

Pathway Analysis of RNA-seq Data: Reducing Redundancy and False Positives

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Abstract

Pathway enrichment analysis is widely used to interpret RNA-seq gene expression data, but it often produces long and redundant pathway lists due to overlapping gene sets and gene-gene correlation. These issues complicate biological interpretation and can inflate pathway-level significance.

The aim of this project is to develop and compare pathway analysis pipelines that reduce redundancy and false positives while preserving biologically meaningful signal. Two pipelines were implemented using the same RNA-seq dataset and background definition, but differing in their enrichment strategy and redundancy reduction.

Pipeline A follows a traditional overrepresentation-based approach, while Pipeline B combines a correlation-aware competitive gene set test with a ranked, cutoff-free enrichment analysis and agreement-based filtering. The results indicate that explicitly accounting for gene-gene correlation and pathway redundancy leads to more compact and interpretable pathway enrichment results.

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1 Introduction

RNA sequencing (RNA-seq) has become a standard technique for measuring genome-wide gene expression and is widely used to study molecular changes associated with disease, development, and experimental conditions. A typical RNA-seq analysis produces gene-level statistics that describe differences in expression between experimental conditions. While these results are informative, long lists of differentially expressed genes are often difficult to interpret directly in a biological context.

To address this, pathway enrichment analysis is commonly applied. Pathway-based methods aim to summarise gene-level changes in terms of known biological processes or pathways by testing whether genes associated with a given pathway show coordinated changes. This can substantially improve interpretability by shifting the focus from individual genes to higher-level biological mechanisms. However, pathway enrichment analysis also introduces several well-known challenges.

One major issue is redundancy. Many biological pathways share genes or describe closely related processes, which often leads to long lists of highly overlapping and difficult-to-interpret results. In addition, many enrichment methods implicitly assume that genes are independent, even though genes within the same pathway are frequently co-expressed. Ignoring gene-gene correlation can inflate pathway-level significance and increase the number of false positive findings [6]. As a result, enrichment analyses may produce results that are statistically significant, but biologically uninformative or misleading. [7]

The goal of this project is to investigate how these issues can be addressed through careful methodological design. In particular, this work focuses on developing and comparing two pathway analysis pipelines that aim to reduce redundancy and false positives while preserving biologically meaningful signal. Both pipelines operate on the same RNA-seq dataset and use the same background definition, but differ in their enrichment strategy and in how pathway-level results are filtered and summarised.

Pipeline A follows a more traditional over-representation-based approach combined with redundancy reduction techniques, while Pipeline B emphasises robustness by using correlation-aware and ranked enrichment methods together with agreement-based filtering. By implementing these two pipelines and analysing their behaviour, this project aims to better understand how different methodological choices affect the size, stability, and interpretability of pathway enrichment results.

1.1 Research Questions

The project is guided by the following research questions:

1. Which pipeline makes the most concise list while maintaining high differentially expressed (DE) gene coverage?
2. Does limiting the background to measured genes reduce false positives?
3. Does bootstrap stability (resampling) improve reproducibility without a substantial loss of coverage?

2 Literature Review

Pathway enrichment analysis is widely used to interpret high-throughput gene expression data by aggregating gene-level signals into biologically meaningful processes. Classical over-representation analysis (ORA) tests whether predefined gene sets are enriched among differentially expressed genes using a hypergeometric or Fisher’s exact test [4]. While computationally simple, ORA depends on an explicit

significance cutoff and frequently produces long, overlapping lists of pathways, limiting interpretability [7].

To reduce sensitivity to hard cutoffs, ranked gene set enrichment methods such as Gene Set Enrichment Analysis (GSEA) evaluate coordinated shifts in pathway genes across an ordered gene list [12]. Ranked approaches are more sensitive to subtle expression changes, but still rely on assumptions about gene independence. Since genes within pathways are often correlated, ignoring this structure can inflate pathway-level significance and increase false positives.

Correlation-aware competitive gene set tests address this limitation by explicitly modelling inter-gene correlation. CAMERA is a representative example that yields more conservative and statistically reliable enrichment results compared to classical ORA [14]. Recent methodological reviews emphasize that accounting for gene–gene correlation, background definition, and multiple testing assumptions is critical for robust pathway analysis [8, 6].

Beyond statistical significance, redundancy caused by overlapping gene sets and hierarchical pathway structures poses a major interpretability challenge. In Gene Ontology (GO), many enriched terms describe closely related biological processes, leading to near-duplicate results. Proposed solutions include GO-graph decorrelation [3], semantic similarity measures [13], and overlap-based network representations of enrichment results [10]. These approaches aim to summarise enrichment outputs into compact, interpretable pathway sets while preserving biological signal.

Taken together, the literature suggests that no single enrichment method adequately balances sensitivity, robustness, and interpretability. Instead, combining complementary enrichment strategies with explicit redundancy reduction provides a more reliable framework for pathway-level interpretation, motivating the design of the two analysis pipelines developed in this project.

3 Data

This project is based on an RNA-sequencing (RNA-seq) dataset combined with curated biological pathway annotations. The gene expression data consist of a processed expression matrix containing normalized expression values for 17,453 genes measured across 8 samples. Genes are identified using official gene symbols (SYMBOL). In addition, gene-level statistical results from a differential expression analysis were used to define genes of interest for downstream pathway analysis.

Pathway annotations were obtained from two widely used biological knowledge bases: WikiPathways [1] and Gene Ontology (GO) [2]. WikiPathways represents curated biological pathways as sets of participating genes, while Gene Ontology provides a structured vocabulary describing biological processes. Gene-pathway relationships were used to link measured genes to pathway definitions.

Because not all genes measured in the RNA-seq experiment are present in pathway databases, and not all annotated pathway genes are measured in a given dataset, a gene universe was defined. The universe consists of genes that are both measured in the expression data and present in the pathway annotations. In this dataset, 6,992 genes satisfied this criterion and were used consistently as the background for all pathway enrichment analyses. This restriction avoids inflated enrichment significance caused by genes that were not observable in the experiment.

4 Methods

This research develops and compares two pathway-analysis pipelines for RNA-seq gene expression data. The goal of both pipelines is to identify biological processes that show coordinated changes between two experimental conditions, while addressing common challenges in pathway analysis such as gene-gene correlation, pathway redundancy, and inflated significance due to overlapping gene sets [4, 7, 8].

Both pipelines use the same input data and background definition, but differ in their statistical testing strategy and in how pathway-level results are filtered

and summarised. By implementing and comparing these pipelines, this work aims to evaluate how different methodological choices affect the robustness and interpretability of pathway analysis results.

4.1 Pipeline A

Pipeline A follows a trust-first pathway analysis strategy. It starts from standard over-representation analysis (ORA) using a measured-gene background and then applies multiple filtering and reduction steps to address redundancy and improve robustness. The main idea is to gradually narrow down pathway results by combining statistical significance, redundancy reduction, and stability across resampling [4, 7].

An overview of the Pipeline A workflow is shown in Figure 1.



Figure 1: Overview of Pipeline A.

4.1.1 Differential expression gene list

To preserve pathway-level signal, a liberal differential expression threshold was used. Genes with raw p -values below 0.05 were selected from the gene-level statistics and intersected with the gene universe. This list was used as the input for over-representation analysis [7].

As a robustness check, the sensitivity of the differential expression gene list to small statistical changes was assessed using a jitter-based approach. Gaussian noise was added to gene-level p -values with standard deviation $\sigma = 0.002$ (very small perturbation) and $\sigma = 0.02$ (stress test affecting borderline genes). The same selection rule was applied, and all resulting DE lists were intersected with the measured gene universe before downstream analysis.

4.1.2 Over-representation analysis

Over-representation analysis was used as the initial enrichment step. For each pathway, a one-sided Fisher’s exact (hypergeometric) test was performed using the measured-gene universe as background,

followed by Benjamini–Hochberg correction across pathways. ORA was applied to both WikiPathways and Gene Ontology Biological Process collections [4, 7].

To improve result quality beyond statistical significance, additional practical filters were applied. Pathway size was restricted to a data-driven range within the universe, a minimum overlap of three differentially expressed genes was required, and an enrichment-over-expectation criterion ($k/\mathbb{E}[k] \geq 1.5$) was used to discourage very large pathways with weak proportional overlap.

4.1.3 GO redundancy reduction

GO Biological Process results are highly redundant due to the hierarchical structure of the ontology. To reduce near-duplicate terms, semantic similarity-based collapse was applied using the Wang semantic similarity measure [13].

GO terms with semantic similarity above a fixed threshold were grouped, while gene membership within pathways was unchanged, and a greedy representative selection was performed, prioritising lower adjusted p-values and stronger overlap metrics. A similarity threshold of 0.60 was selected based on testing several reasonable values to balance redundancy reduction and gene coverage. A hierarchy-based trimming approach was explored as a sensitivity check, but was not used in the main pipeline [3].

The structure of semantic similarity among retained GO terms and the resulting clusters is visualised in Appendix Figure 7.

4.1.4 Overlap-based clustering

Residual redundancy was further reduced using gene-set overlap clustering, applied separately within GO [2] and WikiPathways [1] collections. Gene sets were restricted to the universe and filtered to moderate sizes. A term-term overlap graph was constructed based on Jaccard similarity ($\text{Jaccard} \geq 0.10$) and shared gene counts (at least three genes), and Louvain community detection was used to identify clusters. One representative per cluster was retained using a fixed ranking rule prioritising lower adjusted

p-values and higher gene overlap [10, 5].

A network-based representation of overlap clustering is shown in Appendix Figure 8.

4.1.5 Bootstrap stability and consensus filtering

To assess robustness, bootstrap stability analysis was applied as a separate filtering step on the reduced pathway set. In each iteration, the gene universe was subsampled (fraction $p = 0.85$) and 10% of DE genes were randomly removed as an additional perturbation ($\Delta = 0.10$). ORA was rerun and pathways were marked as selected if they remained significant after correction.

Stability was defined as the selection frequency across iterations and the pathways with stability above a fixed threshold ($\tau \geq 0.70$) were retained.

As a final confidence filter, cross-collection consensus was required between GO and WikiPathways. No collapsing was performed across collections; instead, consensus was defined by sufficient gene overlap between stable GO and WikiPathways representatives. Pathways that were both stable and supported by at least one cross-collection match were retained as the final Pipeline A shortlist [6, 8].

4.2 Pipeline B

Pipeline B is designed to prioritise robust pathway-level signals by combining complementary enrichment strategies and explicit redundancy control. The pipeline integrates a correlation-aware competitive gene set test with a ranked, cutoff-free enrichment analysis, followed by agreement-based filtering and pathway redundancy reduction. The goal is to reduce false positives caused by gene-gene correlation and overlapping pathway definitions, while retaining biologically coherent signals [14, 12, 6].

An overview of the Pipeline B workflow is shown in Figure 2.



Figure 2: Pipeline B workflow.

4.2.1 Correlation-aware competitive pathway testing (CAMERA)

Pipeline B uses CAMERA (Correlation Adjusted MEan RAnk gene-set test) [14] as the primary pathway enrichment method. CAMERA is a competitive gene set test, meaning that for each pathway it compares gene-level differential expression evidence inside the pathway to that of genes outside the pathway, within a predefined background of measured genes.

A key feature of CAMERA is that it explicitly accounts for inter-gene correlation within pathways. This adjustment is important because genes involved in the same biological process are often co-expressed, which can otherwise lead to inflated pathway-level significance [14, 7] if gene independence is assumed. By incorporating correlation estimates, CAMERA provides a more conservative and statistically reliable assessment of pathway enrichment.

CAMERA operates on gene-level statistics obtained from a linear model fitted to the expression data. Pathways were considered significant at a false discovery rate (FDR) threshold of 0.05. These statistics are aggregated at the pathway level while adjusting for within-set correlation, resulting in pathway-specific test statistics and significance values.

Directionality. CAMERA also reports the direction of enrichment, reflecting which experimental condition shows higher expression for genes within a pathway based on the sign of the underlying gene-level statistics.

For all next steps in Pipeline B, pathways enriched in opposite directions were handled independently and never merged during redundancy reduction.

4.2.2 Ranked gene set enrichment analysis (GSEA)

To complement the competitive testing performed by CAMERA, Pipeline B applies a ranked gene set enrichment analysis using fgsea [12]. Unlike CAMERA, gsea does not rely on a predefined differential expression threshold. Instead, it evaluates whether genes belonging to a pathway tend to be consistently enriched toward the top or bottom of a ranked list of

all genes.

Gene-level statistics from the same linear model are used to rank genes. For each pathway, fgsea computes an enrichment score by assessing the distribution of pathway genes along the ranked list. This score is normalised to account for pathway size, producing a normalised enrichment score (NES). Statistical significance is assessed using permutation-based testing with multiple testing correction across pathways, using an adjusted p -value threshold of 0.05. [12, 6]

Within Pipeline B, fgsea serves as a sensitivity analysis that captures subtle, but coordinated expression shifts that may not be detected by threshold-based approaches. Its role is to provide an independent enrichment perspective that complements the correlation-adjusted results from CAMERA.

4.2.3 Agreement-based pathway selection

Following enrichment analysis with CAMERA and fgsea, Pipeline B applies an agreement-based filtering step. The purpose of this step is to prioritise pathways that are supported by both enrichment methods and therefore less likely to reflect method-specific effects.

Agreement required significance in both methods at $\text{FDR} \leq 0.05$ and consistent enrichment direction.

Pathways are retained only if they are supported by both CAMERA and fgsea and show consistent enrichment direction across methods. Enforcing directional agreement ensures that pathways enriched in opposite directions are not combined, which is important for downstream redundancy reduction and interpretation.

This agreement-first strategy reduces the number of candidate pathways while retaining those with converging evidence from complementary statistical approaches [6, 8].

4.2.4 Redundancy reduction and clustering

To improve interpretability, Pipeline B applies additional steps to reduce redundancy among enriched pathways. For Gene Ontology Biological Process terms, semantic similarity is used to merge closely related terms into representative processes based on

shared ontology structure. This helps to limit redundancy caused by hierarchical and synonymous GO annotations. [13]

After semantic collapse, pathways from both Gene Ontology [2] and WikiPathways [9, 1] collections are further grouped using overlap-based clustering, which relies on shared gene membership. Pathways are processed separately by enrichment direction. An overlap graph is constructed in which nodes represent pathways and edges represent substantial gene overlap. For Pipeline B, edges were defined using Jaccard similarity > 0.15 and at least 5 shared genes, and clustering was performed separately for UP and DOWN pathways.

For each cluster, a single representative pathway is selected using predefined criteria prioritising lower adjusted p-values, higher enrichment scores (NES), and more specific pathway definitions. This results in a compact, non-redundant pathway set suitable for downstream interpretation. [10, 5]

5 Results

This section presents the results obtained from the two pathway analysis pipelines. First, the results of Pipeline A and Pipeline B are described separately. Next, the pipelines are compared in terms of pathway redundancy, robustness, and interpretability.

5.1 Pipeline A

Pipeline A follows a traditional over-representation analysis framework combined with structured redundancy reduction and stability assessment. The goal is to retain a broad but robust enrichment signal while reducing redundancy across pathway collections.

5.1.1 Initial enrichment output

Using over-representation analysis on the measured-gene universe, a large number of enriched pathways were identified across both GO Biological Process and WikiPathways collections. After applying statistical and size-based filters ($FDR \leq 0.05$, minimum overlap $k \geq 3$, and enrichment ratio $k/E[k] \geq 1.5$), 292 GO:BP terms and 40 WikiPathways remained.

This initial output reflects a broad enrichment signal capturing many biologically related processes, but with substantial redundancy due to overlapping gene sets and hierarchical pathway definitions.

5.1.2 Redundancy reduction

To reduce redundancy among enriched GO terms, a semantic similarity-based collapse was applied using the Wang measure. At a similarity threshold of $\tau = 0.60$, the 292 GO:BP terms were reduced to 132 representative terms while retaining 97.1% of differentially expressed genes.

Subsequent overlap-based clustering using gene-set overlap further reduced the pathway lists to 53 GO representatives and 32 WikiPathways representatives. These steps substantially reduced redundancy while preserving coverage of the underlying gene-level signal.

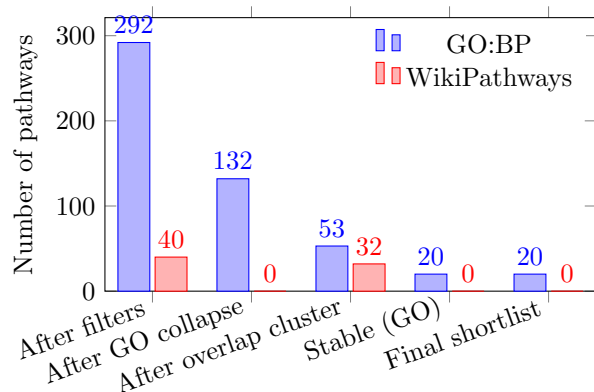


Figure 3: Pipeline A: pathway counts across major reduction steps (values from Section 5.1).

5.1.3 Stability and consensus

Bootstrap resampling ($B = 200$) was applied to assess robustness of the reduced pathway sets. Using a stability threshold of $\tau \geq 0.70$, 20 GO pathways were identified as stable across resampled datasets.

Requiring consensus between GO and WikiPathways based on gene overlap (Jaccard ≥ 0.10 , shared genes ≥ 3) resulted in a final Tier 1 shortlist of 20 pathways. This consensus-based selection prioritizes pathways that are both statistically stable and supported across collections.

5.1.4 Biological characteristics

The final Pipeline A shortlist is dominated by immune, inflammatory, and complement-related pathways, with a smaller subset related to neuronal and synaptic processes. This pattern indicates strong signal retention but limited specificity to Rett syndrome-related neurobiological mechanisms.

A representative subset of the final Pipeline A shortlist is shown in Appendix A (Table 2).

5.2 Pipeline B

Pipeline B applies a correlation-aware pathway analysis framework combined with ranked enrichment analysis and structured redundancy reduction. The goal is to obtain a compact and robust pathway shortlist while reducing false positives caused by gene-gene correlation and overlapping pathway definitions.

5.2.1 Pathway enrichment and agreement filtering

Using the measured-gene universe, CAMERA was applied to 699 WikiPathways and 1,841 Gene Ontology Biological Process gene sets. In parallel, fgsea was run on the same collections using the full ranked gene list.

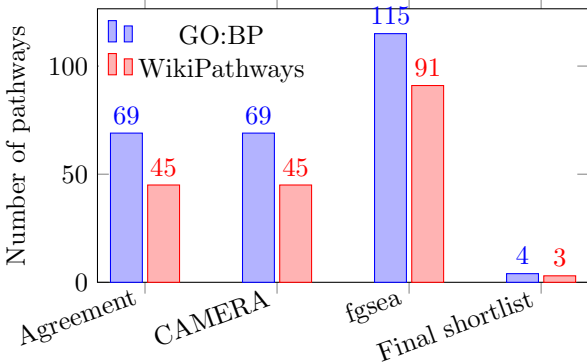


Figure 4: Pipeline B: pathway counts across steps. CAMERA significant at $FDR \leq 0.05$; fgsea significant at adjusted $p \leq 0.05$; agreement requires significance in both and consistent direction.

As expected, fgsea returned a larger number of enriched pathways, reflecting its sensitivity to coordinated but subtle expression shifts. Requiring agree-

ment between CAMERA and fgsea, defined as significance in both methods ($FDR < 0.05$) and consistent enrichment direction, substantially reduced the candidate set to 45 WikiPathways and 69 GO Biological Process terms.

In this dataset, the set of pathways retained after agreement filtering was very similar in size to the set identified by CAMERA alone (Figure 4). This indicates that most CAMERA-significant pathways were also supported by fgsea. In Pipeline B, fgsea therefore primarily serves as an independent confirmation step and provides ranked enrichment information used during redundancy reduction rather than substantially altering the final pathway list.

5.2.2 Redundancy reduction

Redundancy reduction was applied to the agreement-filtered pathways. For GO Biological Process terms, semantic similarity-based collapsing using the Wang measure with threshold $\tau = 0.60$ reduced the set to approximately 50 representative terms.

Following semantic collapse, both GO and WikiPathways results were subjected to overlap-based clustering using shared gene overlap (Jaccard index > 0.15 , minimum of 5 shared genes). Pathways were processed separately by enrichment direction to avoid biologically invalid merges. Most pathways formed single-member clusters, indicating that agreement filtering had already removed much of the redundancy.

5.2.3 Sensitivity analysis of final selection

To assess sensitivity to threshold choices, two consensus configurations were compared. A strict configuration using higher overlap thresholds produced a final shortlist of 7 pathways, while a relaxed configuration yielded a larger set of 27 pathways.

The strict configuration was selected for reporting, as it produced a compact and stable shortlist supported by consistent evidence across methods.

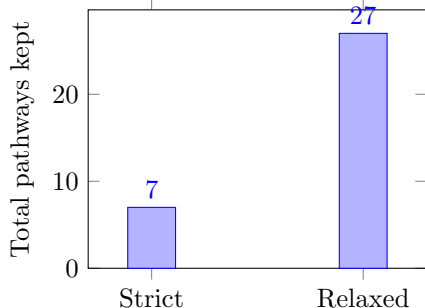


Figure 5: Sensitivity analysis: final shortlist size under strict vs. relaxed final selection.

5.2.4 Final Pipeline B results

The final Pipeline B shortlist consists of 7 high-confidence pathways, including 4 Gene Ontology Biological Process terms and 3 WikiPathways. These pathways were consistently supported by correlation-adjusted testing, ranked enrichment analysis, and redundancy reduction.

Overall, Pipeline B reduces a large and potentially redundant enrichment output to a compact shortlist that prioritizes specificity, stability, and interpretability. A summary of the final pathways is provided in Table 3.

5.3 Comparison of Pipelines

To compare the impact of different pathway analysis strategies, the results of Pipeline A and Pipeline B were evaluated using the same RNA-seq dataset and gene universe. The comparison focuses on pathway compactness, redundancy reduction, and biological specificity.

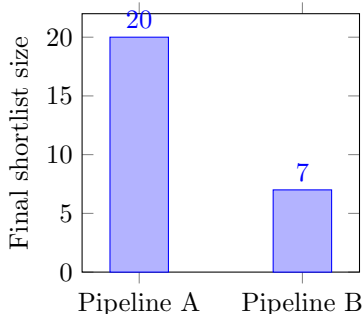


Figure 6: Comparison of final shortlist sizes produced by Pipeline A and Pipeline B.

Pipeline A produced a broader final shortlist consisting of 20 pathways after stability and consensus filtering. These pathways were dominated by immune, inflammatory, and complement-related processes, with a smaller subset related to neuronal and synaptic functions. This reflects strong signal retention but limited specificity to Rett-related neurobiological mechanisms.

In contrast, Pipeline B yielded a substantially smaller final shortlist of 7 pathways. By combining correlation-aware testing, ranked enrichment analysis, and agreement-based filtering, Pipeline B removed many generic and method-specific signals. The resulting pathways showed a stronger focus on neuronal, synaptic, and regulatory processes, while reducing the dominance of broad immune-related pathways.

Overall, Pipeline A prioritizes robustness and signal retention, producing a wider set of stable pathways, whereas Pipeline B prioritizes specificity and conservative inference, resulting in a compact and more interpretable shortlist. The two pipelines therefore represent complementary analytical strategies rather than competing approaches.

A direct comparison of the final pathway shortlists produced by Pipeline A and Pipeline B is provided in Table 1, highlighting differences in compactness and database composition.

	Pipeline A	Pipeline B
Final shortlist size	20	7
GO:BP in final shortlist	20	4
WikiPathways in final shortlist	0	3

Table 1: High-level summary of final outputs.

6 Discussion

This section discusses the results of the two pathway analysis pipelines with a focus on methodological behavior rather than detailed biological interpretation. The aim is to reflect on how different enrichment strategies and filtering choices affect redundancy, robustness, and interpretability of pathway analysis results.

6.1 Pipeline A

Pipeline A follows a trust-first design in which an intentionally permissive over-representation analysis is used to retain a broad enrichment signal before applying redundancy reduction and stability filtering. As a result, the initial enrichment output contained a large number of pathways, particularly from the GO Biological Process collection, reflecting the sensitivity of ORA to overlapping gene sets.

Subsequent semantic collapsing and overlap-based clustering substantially reduced redundancy while preserving most differentially expressed gene coverage. The final stability and consensus filtering further reduced the pathway list to a moderate-sized shortlist. This approach prioritizes signal retention and robustness, but also results in a final set that remains dominated by broad immune and inflammatory processes. While these pathways are biologically relevant, their general nature limits specificity to Rett-related neurobiological mechanisms.

Overall, Pipeline A produces a stable and comprehensive pathway shortlist, but at the cost of reduced biological specificity.

6.2 Pipeline B

Pipeline B was designed to address limitations of traditional enrichment methods by explicitly accounting for gene-gene correlation and avoiding hard differential expression cutoffs. The use of CAMERA provided a more conservative enrichment signal by correcting for within-pathway correlation, while ranked enrichment analysis using fgsea enabled detection of coordinated but subtle expression shifts across the full gene list.

Agreement-based filtering between CAMERA and fgsea substantially reduced the number of candidate pathways, retaining only those supported by independent enrichment perspectives and consistent directionality. This step removed many method-specific and unstable signals. Redundancy reduction was then applied within enrichment direction, further improving interpretability without substantially inflating the final pathway count.

As a result, Pipeline B produced a compact and

stable shortlist with a clearer focus on neuronal, synaptic, and regulatory processes. This increased specificity comes at the cost of reduced sensitivity to broad disease-related pathways, reflecting a more conservative analytical strategy.

6.3 Comparison of Pipelines

Comparing the two pipelines highlights a trade-off between signal retention and specificity. Pipeline A retains a broader set of enriched pathways and emphasizes robustness through stability filtering, but remains susceptible to redundancy and dominance of general biological processes. In contrast, Pipeline B applies stricter filtering and correlation-aware testing, resulting in a smaller and more interpretable shortlist.

The two pipelines therefore represent complementary analytical strategies rather than competing approaches. Pipeline A may be more suitable for exploratory analyses where sensitivity is prioritized, while Pipeline B is better suited for generating concise, high-confidence pathway summaries. More generally, this comparison illustrates how accounting for gene-gene correlation and pathway redundancy can substantially improve the interpretability of pathway enrichment results.

7 Conclusion

This project examined how different pathway enrichment strategies influence the redundancy, robustness, and interpretability of RNA-seq pathway analysis results. Two pipelines were implemented and compared using the same dataset but differing in their enrichment methodology and filtering strategy.

Pipeline A followed a traditional over-representation-based approach and retained a broader set of enriched pathways, demonstrating strong signal retention but limited specificity. In contrast, Pipeline B incorporated correlation-aware and ranked enrichment methods combined with agreement-based filtering, resulting in a more compact and biologically focused pathway shortlist.

Overall, the comparison highlights that explicitly

addressing gene–gene correlation and pathway redundancy can substantially improve the interpretability of pathway enrichment results. These findings emphasize the importance of methodological choices in pathway analysis and motivate the use of more conservative and structured approaches in exploratory transcriptomic studies.

8 Future Work

Future work will focus on further refining both pipelines through additional experiments and methodological variations, such as exploring alternative threshold settings and enrichment configurations. The pipelines will also be applied to other RNA-seq datasets to assess whether the observed behavior generalizes beyond the current dataset and pathway databases. This will help determine the robustness and broader applicability of the proposed approaches.

9 Self-reflection

This project was my first experience working on a longer and less structured research task, which required independent planning, iteration, and decision-making over an extended period of time. Overall, the semester went well, but it also highlighted several areas where I would like to improve.

One aspect that went well was the gradual development of the analysis pipelines. Starting from simple enrichment methods and iteratively adding robustness checks and redundancy reduction helped me better understand how methodological choices affect results. I also gained experience in structuring a research project, managing intermediate outputs, and keeping track of design decisions throughout the semester.

At the same time, this project showed me that I need to improve my time management and planning for larger assignments. Some parts of the analysis and writing were left until relatively late, which made the final phase more stressful than necessary. I also realised that I should spend more time early on defining

clear intermediate goals and writing partial drafts, even when results are not yet final.

Overall, this project helped me better understand how research-oriented data analysis differs from standard coursework. In the next semester, I aim to work more consistently over time, document decisions more clearly as I go, and become more confident in balancing exploration with structured reporting.

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A Appendix

ID	Pathway description	Database	Category
WP4875	Disruption of postsynaptic signaling by CNV	WikiPathways	Neuronal
WP5090	Complement system in neuronal development and plasticity	WikiPathways	Neuro-immune
WP3591	Sleep regulation	WikiPathways	Physiological
WP75	Toll-like receptor signaling	WikiPathways	Immune
GO:0006954	Inflammatory response	GO:BP	Immune
GO:0006955	Immune response	GO:BP	Immune

Table 2: Representative subset of the final Pipeline A shortlist. The full shortlist contains 20 pathways and is summarised here to illustrate the breadth of biological processes retained by Pipeline A.

ID	Pathway	Database	Dir	CAMERA_FDR	fgsea_padj	agreement_q	set_size
WP4875	Disruption of postsynaptic signaling by CNV	WikiPathways	UP	0.002379	0.002959	0.002959	33
GO:0035418	protein localization to synapse	GO:BP	UP	0.01137	0.0217	0.0217	16
GO:0048168	regulation of neuronal synaptic plasticity	GO:BP	UP	0.03634	0.02514	0.03634	21
WP2328	Allograft rejection	WikiPathways	DOWN	0.0002474	2.284e-06	0.0002474	54
WP5491	Transcriptional regulation of memory B cell differentiation	WikiPathways	DOWN	0.00512	0.004276	0.00512	33
GO:0019886	antigen processing and presentation of exogenous peptide antigen via MHC class II	GO:BP	DOWN	0.008493	0.009154	0.009154	20
GO:0002224	toll-like receptor signaling pathway	GO:BP	DOWN	0.002644	0.01922	0.01922	21

Table 3: Final Pipeline B shortlist (7 pathways).

